Name:_____ Student ID:_____

CSE 4074 Computer Networks

Fall 2023 Final Exam - 22.01.2023 Monday

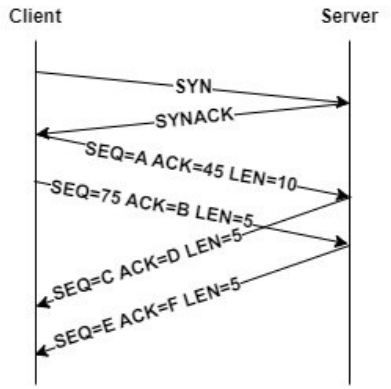
Duration: 40 Minutes for Part 1, Total: 105 Minutes.

Q1	Q2	Q3	Q4	Q5	Q6	SUM
/45	/10	/15	/24	/8	/23	/100

Group-A

Q-1. (48 pts) Multiple choice questions.

Consider the following TCP flow between a client and a server. There is no timeout event. Answer questions 1-3.



1) What are the values of A, B and C?
   a. 65, 45, 45
   b. 65, 50, 45
   c. 65, 55, 55
   d. 55, 55, 65
   e. 65, 45, 75

2) What are the values of D, E and F?
   a. 55, 80, 60
   b. 55, 60, 65
   c. 75, 50, 80
   d. 75, 60, 80
   e. 80, 60, 60

3) Which of the following is **not true**?
   a. Initial SEQ number of the server is learned by the client via the SYNACK message.
   b. Third step of the 3-way handshaking contains client to server data.
   c. In the SYNACK segment, both the SYN bit and ACK bit is set to 1.
   d. If the segment with ACK=D is not received by the client successfully due to bit errors, and all the other segments are successfully received, the client should retransmit the segment with SEQ=A.
   e. If the segment with SEQ=A is not received successfully due to bit errors, the value of F would be different compared to the no-error case.

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4) Consider that a Go-Back-N sender sends four packets. None of the packets are lost, but all the ACKs are lost except the ACK of the third packet. Timeout event occurs at the sender after receiving the ACK of the third packet. Which packet(s) would be retransmitted by the sender?

a. The first two packets.
b. The first three packets.
c. The first, the second, and the fourth packets.
d. Only the fourth packet.
e. All the four packets.

5) Suppose that EstimatedRTT for a TCP sender is initially 100ms, and exponential weighted moving average is used (with alpha=0.125) to update this value. The next sample RTTs are A, B, C and D milliseconds respectively. If the final value of the EstimatedRTT is greater than 100 ms, which of the following is not a possible value of A, B, C, and D?

a. A=100, B=100, C=90 D=110
b. A=90, B=90, C=90 D=130
c. A=110, B=110, C=110, D=98
d. A=90, B=110, C=90, D=110
e. A=120, B=80, C=100, D=100.

6) Which of the following header fields are common in both TCP and UDP segments?

I. Source and destination ports
II. Checksum
III. SEQ and ACK numbers
IV. Cwnd (congestion window)

a. Only I
b. Only II
c. I and II
d. I, II and III
e. I, II and IV

7) Which of the following is **false** about TCP congestion control? (MSS= Maximum Segment Size)

a. Congestion window is set to 1 MSS initially.
b. In slow start, TCP congestion window increases exponentially until a threshold value.
c. In slow start, congestion window is incremented by 1 MSS for every ACK received.
d. After the threshold value, congestion window is increased linearly, i.e. in additive manner.
e. When a packet is lost, the threshold value is set to half of the previous threshold value.

8) Which of the following TCP congestion control protocols is most popular in popular web servers, and provides higher throughput in long links with large bandwidth?

a. TCP Tahoe
b. TCP Reno
c. TCP New Reno
d. TCP CUBIC
e. TCP Vegas

9) Which protocol is used to learn the IP address of the first hop router?

a. ICMP
b. DHCP
c. SNMP
d. ARP
e. FHRP

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10) Suppose that IP address of host A is 192.167.2.136 and subnet mask is 22. Which of the following is a possible IP address of host B that is in the same subnet?

I. 192.167.1.156
II. 192.167.2.199
III. 192.167.3.136
IV. 192.167.5.135

a. Only II
b. I and II
c. II and III
d. I, II and III
e. II, III and IV

11) Which of the following are new features added to IPv6 to decrease the processing delay compared to IPv4?

I. Fragmentation is not supported when packets are too big.

II. Header size is fixed.

III. There is no checksum for error detection.

a. Only II
b. Only III
c. I and II
d. II and III
e. I, II and III

12) In a router, which is the following can be a reason for packet queueing and loss?

I. Packets arrive to the input port very frequently
II. Switching fabric is slow
III. Transmission delay in the output link is large.

a. Only I
b. Only II
c. I and II
d. I and III
e. I, II and III

13) Which of the following is **not true** about generalized forwarding?

a. It is possible to configure a router as a firewall.
b. It is possible to configure a router as a link-layer switch.
c. Port number of a segment carried by an incoming IP packet cannot be changed.
d. It is possible to forward TCP segments and UDP segments with the same source-destination pair to different output ports.
e. It is possible to forward HTTP packets and SMTP packets with the same source-destination pair to different output ports.

14) If data is 100101 and the generator is 1011 what would be the CRC bits?

a. 010
b. 110
c. 101
d. 0010
e. 0110

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15) Which of the following is not correct for Medium Access Control protocols?

a. In CSMA/CD, collision possibility decreases when propagation delay increases.
b. In high load, such as wireless wide area networks, it is better to use channel partitioning protocols
c. In low load, such as in home WiFi network, it is better to use random access protocols.
d. In Bluetooth, nodes with more to send can take longer turns.
e. In Cable network, channel partitioning and random access protocols are used together.

16) In SDN, some messages and a potential use of these messages. Which of them is/are **not correct**?

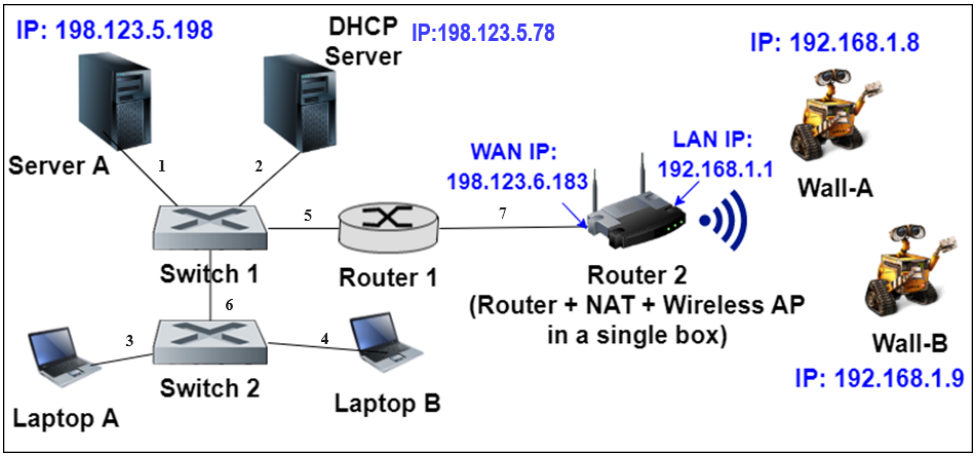
I. Packet in: Forward the packet to the next router.
II. Port-status: Inform the controller when a link cost changes.
III. Modify-state: To add a new entry to the flow table for NAT traversal.

a. Only I
b. Only II
c. Only III
d. I and II
e. I and III

17) Which of the following protocols is used to send ping message to www.unitech.ac.pg which is the web site of Papua New Guinea University of Technology?

a. DHCP
b. ICMP
c. YANG
d. SNMP
e. NETCONF

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Q-2 (41 pts) Consider the above network, where two trash-collecting robots can be remotely controlled. All switches are learning switches. ARP tables and switch tables are initially empty. Router 2 has NAT, Router 1 does not have. Suppose that both Wall-A and Wall-B are running a server process listening the port 5566. There are also client processes of Wall-A and Wall-B which use ports 1234 and 4321 respectively. Server A is running a server process listening the port 8080. You can name MAC addresses of the interfaces as MAC-Server-A, MAC-Laptop-A, etc. Answer the following questions:

(a – 4 pts) Suppose that Wall-B wants to establish a connection with Server A and sends a TCP SYN segment. What entry would be added to the NAT table of Router 2?

(b – 4 pts) Does Wall-B need to know the LAN-IP of Router 2 to send the TCP SYN segment? Why?

(c – 4 pts) Does Router 1 need the MAC address of Server A? If not, why? If yes, why it needs and how it learns?

(d – 4 pts) After all the necessary communication needed to send TCP SYN from Wall-B to Server A, what would be the contents of the switch table in Switch 2? Use the numbers given over the links.

(e – 4 pts) Now suppose that Server A sends back a SYNACK segment. What actions would be taken in Router 2?

(f – 6 pts) How can Laptop A remotely control Wall-A without knowing its IP address? Offer a solution using Server A.

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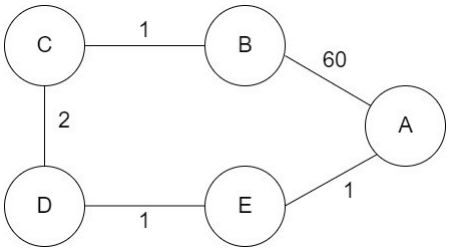
(g – 3 pts) How many subnets are there in the above network?

(h – 4 pts) Suppose that Laptop B is newly connected and wants to get an IP address from the DHCP server. What would be the IP and MAC destination address of the DHCP request?

(i – 4 pts) Show that the subnet mask of Laptop B cannot be /22.

(j – 4 pts) Suppose that Wall-A cannot hear the signals of Wall-B. Is this a problem if they are using CSMA as a multiple access protocol? Why?

Q-2. (24 pts) Consider the following graph:



(a – 8 pts) Show the operation of Dijkstra's (Link State) algorithm for computing the least cost path from **E** to all destinations. How does E set its forwarding table entries for all destinations if the traditional destination-based forwarding is used?

(b – 12 pts) Suppose that Distance Vector Routing is used and all the distance vectors are converged. Show the distances vectors of C and D. Describe how removal of CD link causes count-to-infinity problem.Approximately how many DV messages will be sent by C and D?

(c – 4 pts) In BGP, what could be a reason for using Hot-potato routing instead of Shortest-AS path?